How Much of an Expected Goals Model Is in the Words?

Recovering shot quality from free-text match commentary

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***Abstract*—Expected goals (xG) models are estimated from shot coordinates, which are collected by human annotators or tracking cameras and, outside a small number of open releases, licensed rather than published. Free-text match commentary is available without a key and describes each shot in a single sentence. We ask how much of a coordinate-based xG model survives when that sentence is the only input. Parsing 87,980 shots from ESPN commentary across six European competitions into eighteen fields, a logistic regression attains an AUC of 0.7709 on a held-out Premier League season. On 8,825 shots joined to StatsBomb's open 2015/16 release, for which both a sentence and a coordinate xG exist for the same shot, the text model reaches 0.7826 against 0.8118, recovering 90.6% (95% CI 86.2–95.0) of the coordinate model's discrimination above chance; per team-match the two correlate at $r = 0.869$, within the range reported between commercial providers. Substituting the full sentence for the extracted fields costs 0.010 AUC, so the extraction is not the binding constraint. We further show that text-derived features drift in level but not in ranking. The model over-estimates the held-out season by 11.9%, which we localise to two phrases requiring a judgement by the annotator while unambiguous phrases are unchanged over a decade, and a single in-season intercept refit reduces the error to 2.9% without affecting the ranking.**

***Index Terms*—expected goals, football analytics, text as features, concept drift, calibration**

I. Introduction

An expected goals model estimates the probability that a given shot results in a goal. Every widely used specification is built on the location from which the shot was taken: a coordinate pair, a derived distance and angle, and increasingly the positions of the other players in frame [1], [2]. That location is the expensive component. It is produced either by human annotators or by tracking cameras, and with the exception of a small number of open releases [3] it is licensed rather than published. Robberechts and Davis [4] make the cost explicit from the modelling side, showing how much data an xG model needs before it converges; the constraint we address is prior to theirs, in that the data may not be obtainable at all.

Free-text commentary carries none of that cost. For each shot a sentence is published live and without authentication, of the form shown at the top of Figure 1. Such sentences exist for many competitions for which open coordinate data does not and is unlikely to. They plainly encode part of what an xG model requires, and the question we address is what proportion.

We treat this as a measurement problem rather than a feasibility question, because the practical implication depends on the magnitude. Recovery of a small fraction of a coordinate model would be of limited interest. Recovery of most of one would make xG estimable in any competition for which a commentary feed is written, a substantially wider population of competitions than those with open coordinates.

Contributions

1) We quantify how much of a coordinate xG model is recoverable from commentary text. On 8,825 shots for which both a commentary sentence and a StatsBomb coordinate xG are available, a model given only the sentence recovers 90.6% of the coordinate model's discrimination above chance (Section VI-B).
2) We calibrate that figure against the agreement observed between commercial providers, which is the external scale we found for it (Section VI-C).
3) We establish that the extraction is not the binding constraint. Models given the full sentence, including all 1–4 grams and a sentence embedding, perform worse than the eighteen extracted fields by 0.010 AUC (Section VI-D).
4) We characterise concept drift in text-derived features. The ranking is stable across a decade, replicating [4] on a different feature source, whereas the calibrated level is not. We localise the drift to two phrases, exhibit control phrases that do not move, and give a leak-free in-season correction (Section VII).
5) We release the full pipeline. Collection, parsing, estimation, evaluation and every figure reported here are reproduced by a single script [5].

II. Related Work

a) Coordinate-based xG: Expected goals models have been studied for roughly two decades. Recent work addresses explainability [2], player and position correction [6], and

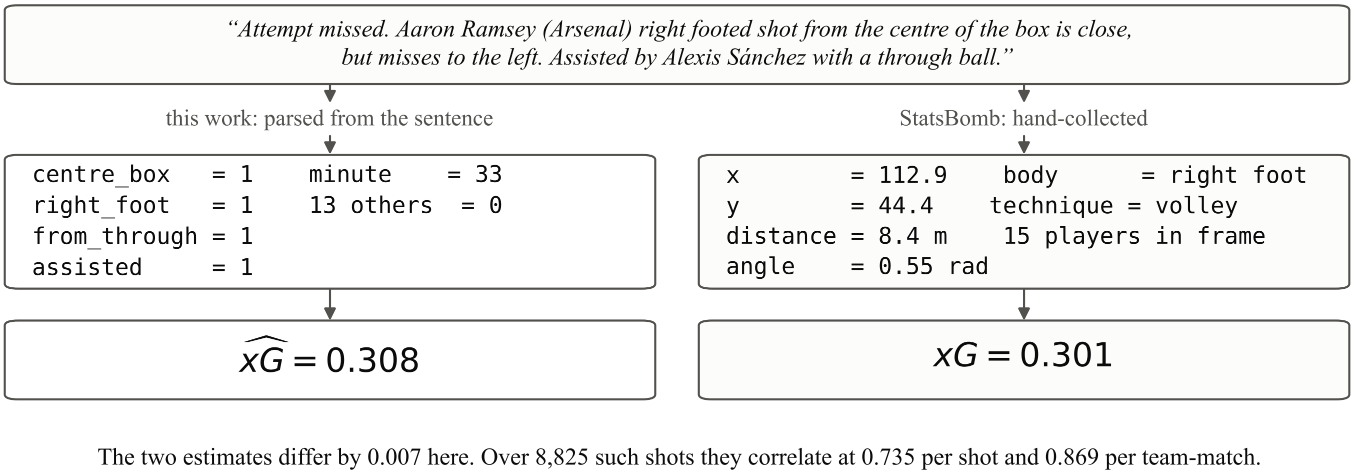

Fig. 1. The same shot as each side of the comparison receives it. The commentary sentence yields eighteen parsed indicators and no geometry; the StatsBomb release yields a location, a derived distance and angle, and the positions of the players in view. Neither side sees the other's representation, and both are scored against the same outcome.

downstream validity [1]. In each case shot location is taken as given.

b) Data requirements: Robberechts and Davis [4] examine what data an xG model requires. They report that a basic logistic specification converges at approximately 6,000 shots, that estimation on less recent data incurs "only a negligible performance hit", that league-specific and cross-league models "perform equally", and that data from different providers should not be pooled. They further argue that calibration rather than ranking is the appropriate objective. Their results bear on the present work in two ways: our staleness and transfer results replicate theirs on a different feature source, and our drift result is interpretable only against the coordinate-side baseline they establish.

c) Language and football: Text has been used to forecast match events from event sequences [7], and to render an xG model's coefficients as natural-language explanations of individual shots [8]. The latter proceeds in the opposite direction to the present work, mapping a fitted model onto words rather than words onto a model. The two are complementary rather than equivalent.

d) Prior use of commentary as input: An unpublished public repository [9] scrapes Premier League match commentary and states that shot position can be inferred from the commentary text, from which goal-conversion rates by shot category are derived. It is the earliest such use we located, and it precedes the present work; our search would not reliably have found an earlier one, so we claim only that the idea is not ours. It reports no fitted model, no held-out evaluation, no comparison against a reference xG model, and no AUC, log loss or Brier score. The contribution claimed here is therefore the measurement rather than the representation.

e) Agreement between providers: Published comparisons of Opta, Understat, StatsBomb and Wyscout report match-level xG correlations between 0.86 and 0.96 depending on the pair [10]. We use these figures as the reference scale for our own agreement with StatsBomb in Section VI-C.

TABLE I
THE COMMENTARY CORPUS. PREMIER LEAGUE SPANS 2015/16 AND 2022/23–2025/26; THE OTHER FIVE COMPETITIONS COVER 2022/23–2025/26.

Competition	Matches	Shots	Goals	Goal rate
Premier League	1,892	47,431	5,489	11.6%
La Liga	380	9,240	1,024	11.1%
Serie A	380	9,065	922	10.2%
Bundesliga	306	7,853	990	12.6%
Ligue 1	305	7,391	863	11.7%
Primeira Liga	306	7,000	821	11.7%
Total	**3,569**	**87,980**	**10,109**	**11.5%**

f) Feature sets in public models: American Soccer Analysis publish their xG specification [11], comprising distance, available goal mouth, whether the shot was headed, whether it followed a cross or through ball, and pattern of play including corners, free kicks, fast breaks and penalties, estimated by logistic regression. Setting aside the two continuous geometric terms, this is close to the field set a commentary sentence yields, which provides independent support for the choice of features in Section IV-B.

III. DATA

A. Commentary

ESPN publish an Opta-derived English commentary feed through a public endpoint that requires no authentication. We collected 3,569 matches across six competitions (Table I), from which 87,980 shots and 10,109 goals were parsed.

B. The reference model

StatsBomb release Premier League 2015/16 as open data [3], including a shot location, a freeze frame of the players in view at the moment of the shot (median 13, range 1 to 19 across the release), and their own xG estimate for each shot. Joining this release to ESPN commentary for the same fixtures puts words and coordinates on identical shots. We use Premier League 2015/16 because ESPN commentary for it was already collected; StatsBomb also release La Liga 2015/16 to 2020/21, Ligue 1 2021/22 and 2022/23, Bundesliga 2023/24, Serie A 2015/16 and Premier League 2003/04, all of which ESPN also cover, so further validation sets are obtainable and this one is not the only possible comparison.

The join is deliberately conservative. A shot is paired only when the fixture, the team and the minute all agree, and an unmatched shot is discarded rather than imputed. This yields 8,825 shots across 373 matches. As a check on the join itself, the two sources agree on whether a shot was a goal for 99.71% of paired shots, disagreeing on 26. All evaluation in Section VI-B takes StatsBomb's goal indicator as the label, so that their model is not penalised for annotation differences in the commentary feed.

IV. METHOD

A. Outcome leakage in the commentary sentence

Commentary states the outcome of a shot before describing the chance that produced it. Every line opens with "Goal!", "Attempt missed", "Attempt saved" or "Attempt blocked", and the outcome is frequently restated in the verb, as in "is saved in the bottom left corner" or "hits the left post". That text constitutes the label. A model estimated on the unmodified sentence attains an AUC of 1.00 while learning nothing about the quality of the chance.

Reliable removal of the outcome proved difficult, and four distinct leaks survived successive attempts:

1) Stripping only the opening clause left penalties converting at 100%.
2) A blacklist of outcome words failed to remove the verbs; inspection of the fitted coefficients found `goal` weighted at +7.99.
3) Under a whitelist, retaining only spans matching a fixed set of chance-describing patterns, the phrase "from a free kick" still converted 60 of 60. Free kicks are consequently excluded from the feature set.
4) The `header` indicator was being set by the phrase "headed *pass*", mislabelling 1,636 shots and raising their apparent conversion from 10.0% to 13.8%. Fields describing the shot itself are now read only from the text preceding "assisted by".

These cases share a structure. A leak in text features is a phrasing that was not anticipated, so a fixed word list cannot bound the problem and the safeguard must be behavioural. We therefore estimate a model on the stripped text and require that it not separate goals beyond a specified threshold, which we set at AUC 0.85 against 1.00 for the unmodified text. We additionally flag any n-gram whose Wilson lower bound on conversion exceeds that of penalties, the highest-converting legitimate feature. This procedure identified leak (3), and correctly admitted a genuine high-conversion phrase, "close range following fast break" at 26 of 32, that a fixed 80% threshold had rejected.

TABLE II
GOAL RATE BY PARSED FIELD, 37,929 PREMIER LEAGUE SHOTS (2022/23–2025/26). OVERALL RATE 11.8%.

Field	% shots	Goals	Field	% shots	Goals
penalty	1.0	82.7%	from cross	18.6	10.7%
six yard box	5.0	39.6%	assisted	75.9	10.7%
fast break	2.7	35.5%	header	18.2	10.5%
through ball	3.5	25.7%	difficult angle	2.5	8.1%
after corner	9.8	16.4%	side of box	18.5	5.6%
centre of box	36.2	14.4%	outside box	30.6	4.1%

B. Features

Eighteen fields are extracted by whitelist regular expression. Six describe the location as a zone: the six-yard box, the centre of the box, the side of the box, outside the box, long range, and a difficult angle. Three give the body part, distinguishing a header from a shot with either foot. Five describe the build-up: a cross, a through ball, a headed pass, a corner, and a fast break. The remaining three record whether the shot was assisted, whether it was a penalty, and the minute of the match. The exact patterns are given in Appendix A.

Table II reports the conversion rate associated with each field. The observed range spans roughly a factor of twenty, from 4.1% for shots described as coming from outside the box to 82.7% for penalties.

C. Specification

The model is a logistic regression on standardised features, estimated with scikit-learn [12]. We adopt this specification rather than a more flexible learner because eighteen predominantly binary fields offer limited scope for interaction effects, a choice Section VI-D supports empirically against gradient-boosted trees [13].

V. EXPERIMENTAL SETUP

A. Splits

Two distinct fits are reported and should be distinguished. The *primary* fit is estimated on Premier League 2022/23–2024/25 (28,735 shots) and evaluated on the held-out 2025/26 season (9,194 shots). The *comparison* fit is estimated on Premier League 2022/23–2025/26 (37,929 shots) and evaluated on 2015/16, which is excluded from its estimation sample; this is the fit evaluated against StatsBomb. Splits are by season rather than at random, so that no shot from a match in an evaluation sample can influence the model through another shot in the same match.

All estimation is restricted to the Premier League, following [4] on pooling across sources. The remaining five competitions are held out entirely and used as a transfer test (Section VI-E).

B. Baselines

We compare against four baselines, each chosen to isolate one of the questions the paper asks.

a) Base rate: A constant prediction at the estimation-sample conversion rate, which fixes AUC at 0.500 by construction and gives the Brier score any informative model must beat.

b) The reference coordinate model: StatsBomb's own published xG for each shot, which uses the location, the derived geometry and the freeze frame. This is the upper reference for the recovery figure, and the only baseline that sees information the commentary sentence does not contain.

c) Readers of the full sentence: Two specifications given the entire stripped sentence rather than the extracted fields: TF-IDF over all 1–4 grams, and a MiniLM sentence embedding [14]. These test whether the extraction, rather than the sentence, is what limits performance. A third variant concatenates the embedding with the extracted fields.

d) A stronger learner on the same fields: Gradient-boosted trees [13] on the eighteen extracted fields, which tests whether the linear specification leaves interaction structure unused.

We do not compare against Opta, Understat or Wyscout, whose xG values are not openly available per shot; their published pairwise agreement is used instead as a reference scale in Section VI-C.

C. Evaluation metrics

We report three quantities, which answer different questions and are not interchangeable.

a) AUC: is the area under the receiver operating characteristic curve [15], equal to the probability that a randomly chosen goal receives a higher estimate than a randomly chosen non-goal. It measures *ranking* only and is invariant to any monotone transformation of the predictions, so a model may attain a high AUC while being badly calibrated. Chance is 0.500.

b) Brier score: [16] is the mean squared difference between the predicted probability and the outcome. Unlike AUC it is minimised only by well-calibrated probabilities, which is why [4] argue it is the appropriate primary metric for xG. Lower is better.

c) Log loss: is the mean negative log-likelihood of the observed outcomes. It is also calibration-sensitive and penalises confident errors more sharply than the Brier score.

We separate ranking from calibration throughout, because the central result of Section VII is that the two behave differently under drift.

d) Recovery: The quantity of interest in Section VI-B is the share of the reference model's discrimination above chance that the commentary model reproduces,

$$\rho = \frac{\mathrm{AUC}_{\text{text}} - 0.5}{\mathrm{AUC}_{\text{coord}} - 0.5}, \tag{1}$$

which is a ratio of two statistics estimated on the same shots and therefore carries sampling error. We obtain a percentile interval by bootstrap [17] over 2,000 resamples, resampling *matches* rather than shots: shots within a match share a fixture, two teams and one writer, so resampling shots independently would understate the variance.

TABLE III
WORDS AGAINST COORDINATES, 8,825 SHOTS, PREMIER LEAGUE 2015/16, LABELLED BY STATSBOMB. METRICS ARE DEFINED IN SECTION V-C; THE BASE RATE IS THE ESTIMATION-SAMPLE CONVERSION RATE, SO ITS AUC IS 0.500 BY CONSTRUCTION.

Model	Input	AUC	Log loss	Brier
Base rate	—	0.5000	0.3283	0.0909
Commentary	sentence	0.7826	0.2711	0.0764
StatsBomb	coordinates	0.8118	0.2555	0.0715

TABLE IV
MATCH-LEVEL XG AGREEMENT. PROVIDER FIGURES FROM [10] (FIVE LEAGUES, FIVE SEASONS, ≈4,290 MATCHES); OURS COMPUTED ON 746 TEAM-MATCHES OF THE JOIN.

Pair	Correlation
Opta × Understat	0.96
Opta × StatsBomb	0.92–0.93
StatsBomb × Understat	0.92–0.93
Wyscout × the others	0.86–0.88
Commentary model × StatsBomb	0.869

VI. RESULTS

A. Held-out season

On the 2025/26 season, which is not used in estimation, the primary model attains an AUC of 0.7709 and a Brier score of 0.0840 against a base rate of 11.3%.

B. Comparison against a coordinate model

Table III reports the central result. Both models are evaluated on the same 8,825 shots, labelled by StatsBomb.

Applying Equation 1,

$$\rho = \frac{0.7826 - 0.5}{0.8118 - 0.5} = 0.906, \quad 95\% \text{ CI } [0.862, 0.950], \tag{2}$$

so the commentary model recovers roughly nine tenths of the coordinate model's discrimination above chance, and the interval is narrow enough to exclude both a small fraction and near-parity. The AUC gap itself is 0.0291 with a 95% interval of [0.0151, 0.0436]; StatsBomb is ahead in every one of the 2,000 match-level resamples, so the gap is real and not an artefact of this sample. What the interval establishes is its size.

The two mean predictions agree to within 0.001, at 0.0972 against 0.0982, so the information the words omit does not enter as a systematic bias. Per-shot correlation is 0.735 (Spearman 0.628), with a mean absolute difference of 0.052. Figure 2 shows that both models are close to calibrated across the range and are difficult to separate on that criterion.

C. Interpreting the recovery figure

A recovery figure of this kind requires an external scale. The natural comparison is the agreement observed between commercial providers, all of which work from coordinates and none of which agrees exactly with the others [10]. Table IV places the commentary model on that scale.

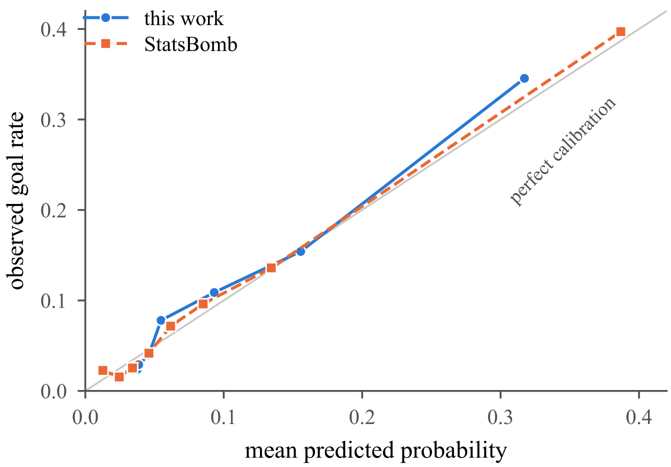

Fig. 2. Reliability of both models on the 8,825 joined shots, in octiles of predicted probability. The two curves lie almost on top of one another and both track the diagonal, so the gap in Table III is a difference in ranking rather than in calibration. Identity is carried by the legend and the marker shapes, since the curves coincide too closely to label in place.

TABLE V
READING MORE OF THE SENTENCE, HELD-OUT 2025/26 SEASON.

Model	Sees	AUC
18 extracted fields, logistic *(primary)*	18 fields	0.7709
18 extracted fields, gradient-boosted trees	18 fields	0.7695
every 1–4 gram, TF-IDF	all words	0.7612
sentence embedding (MiniLM)	all words	0.7584
embedding + extracted fields	both	0.7589

Mean absolute difference is 0.249 xG per team-match, with a median of 0.198 and a maximum of 1.54. Mean xG is 1.150 for the commentary model against 1.161 for StatsBomb and 1.192 observed goals.

A model estimated from English sentences therefore agrees with StatsBomb to approximately the degree that a commercial provider does. Two qualifications apply. First, the levels of aggregation are similar but not identical: the published figures are per match across five leagues and five seasons, whereas ours is per team-match on one league and one season. Second, a frequently repeated figure of approximately 1 xG mean absolute difference between providers, with a maximum of 3.88, is deliberately not used here. The attribution of 3.88 to a single club suggests a season aggregate, which would make the published figure substantially tighter per match than our 0.249 rather than looser, and the underlying source is not openly available, so the unit cannot be confirmed.

D. *Whether extraction is the binding constraint*

The residual 9.4% may arise from either of two sources: the regular expressions capture only anticipated phrasings, or the sentence does not contain the missing information. These can be distinguished by supplying models with the entire sentence and testing whether they improve on the eighteen extracted fields. Table V reports the comparison.

TABLE VI
THE PREMIER LEAGUE FIT APPLIED TO FIVE FURTHER COMPETITIONS WITHOUT RE-ESTIMATION. †THE ESTIMATION LEAGUE.

Competition	Shots	Goal rate	AUC	Mean xG
Premier League†	9,194	11.3%	0.7709	0.127
La Liga	9,240	11.1%	0.7730	0.119
Ligue 1	7,391	11.7%	0.7807	0.122
Bundesliga	7,853	12.6%	0.7795	0.125
Serie A	9,065	10.2%	0.7702	0.115
Primeira Liga	7,000	11.7%	0.7871	0.122

Every specification given the full text performs worse, including the sentence embedding, and appending the embedding to the extracted fields degrades those as well. The remainder of the sentence consists largely of player and team names and connective text, which the models fit as noise. The cost of reading more of the sentence is 0.010 AUC, so no more capable reader, including a large language model used as an extractor, can be expected to close the gap. The residual 9.4% corresponds to information the sentence does not encode: the exact distance within a zone, the shot angle, and the number of intervening defenders.

E. *Transfer across competitions*

The claim that xG becomes estimable wherever commentary is published holds only if the model transfers. We applied the Premier League fit to five further competitions without re-estimation or tuning (Table VI).

Mean AUC across the five held-out competitions is 0.7781 against 0.7709 on the estimation league, a transfer cost of −0.0072 and hence a marginal improvement. Calibration is maintained, with mean predictions of 0.115 to 0.125 against observed rates of 10.2% to 12.6%, and the ordering across competitions is preserved. This replicates the cross-league result of [4] on a feature source they do not consider. A plausible mechanism is specific to this setting: the commentary template is constructed identically across competitions, so the feature distribution is comparatively stable across them.

F. *Error analysis*

Table III gives the size of the gap; this section locates it. Table VII reports the mean absolute difference between the two estimates within each field, restricted to fields appearing on at least 40 of the joined shots.

The ordering is interpretable. Disagreement is smallest where the phrase is geometrically specific: “outside the box” at 0.019 constrains the location tightly enough that little is left for the coordinates to add. It is largest where the phrase is a judgement: “from very close range”, which sets the six-yard-box indicator, disagrees by 0.171, and “following a fast break” by 0.165. Two fields also carry substantial signed bias, with the commentary model reading fast breaks 0.115 and long-range efforts 0.114 higher than StatsBomb does.

Three examples make the mechanism concrete. Where the sentence is specific the estimates coincide to three decimals, as in *“header from the centre of the box assisted by . . . with*

TABLE VII
WHERE THE TWO ESTIMATES DISAGREE, BY PARSED FIELD, ON THE 8,825 JOINED SHOTS. BIAS IS SIGNED, OURS MINUS STATSBOMB'S. FIELDS APPEARING ON FEWER THAN 40 SHOTS ARE OMITTED.

Field	n	mean $\lvert\Delta\rvert$	bias	Field	n	mean $\lvert\Delta\rvert$	bias
six yard box	285	0.171	$+0.000$	difficult angle	194	0.056	$+0.012$
fast break	64	0.165	$+0.115$	header	1,382	0.053	-0.021
through ball	285	0.118	-0.004	penalty	80	0.052	$+0.052$
long range	47	0.114	$+0.114$	assisted	6,771	0.051	-0.004
centre of box	2,896	0.078	-0.005	right foot	4,792	0.051	$+0.003$
after corner	762	0.072	$+0.039$	left foot	2,623	0.050	-0.001
from cross	1,685	0.060	-0.017	side of box	1,493	0.037	-0.013
headed pass	286	0.060	$+0.009$	outside the box	3,511	0.019	$+0.002$

a cross", which both models place at 0.057. Where it is not, the commentary model can be badly wrong in either direction. On *"from very close range …following a corner"* it returns 0.819 against StatsBomb's 0.098, the coordinates placing the shot 8.3 m out; and on *"left footed shot from the centre of the box assisted by"* it returns 0.157 against 0.930, since nothing in that sentence separates a tap-in from a twenty-yard effort within the same zone.

This connects the two halves of the paper. The fields carrying the largest disagreement in Table VII are the same two phrases whose meaning moves across seasons in Section VII. Imprecise phrasing is simultaneously the least accurate feature and the least stable one, which is one mechanism rather than two independent findings: a phrase that admits interpretive latitude will be applied inconsistently within a season and differently between seasons.

VII. CONCEPT DRIFT IN TEXT-DERIVED FEATURES

A. The observed discrepancy

On the held-out season the model predicts 1.54 goals per team-match against 1.38 observed, an over-estimate of 11.9%. This decomposes as follows.

		Factor
league goal rate	$0.1199 \to 0.1134$	$\times\,1.057$
mean prediction	$0.1199 \to 0.1270$	$\times\,1.059$
observed over-estimate		1.119

The first row is the decline in the league's conversion rate between the estimation window and the held-out season. The second requires explanation. The mean prediction of a logistic regression on its own estimation sample equals the sample base rate by construction, so a mean of 0.1270 on 2025/26 implies that the parsed features of those shots are more favourable than those of the shots on which the model was estimated, while the shots themselves produced fewer goals.

B. Localising the shift to individual phrases

Table VIII attributes the second factor to two phrases, and Figure 3 shows the movement season by season. Over the decade separating the two seasons, "following a fast break" became approximately 4.5 times more frequent while its conversion rate fell to a little over half its former value. A change of this magnitude in the underlying football is implausible on both counts simultaneously. The more parsimonious explanation is a change in how loosely the phrase is applied by whoever writes the feed, with the consequence that a model conditioned only on the words attributes higher quality to the shot than was warranted.

TABLE VIII
SHARE OF SHOTS AND CONVERSION RATE FOR FOUR COMMENTARY PHRASES, PREMIER LEAGUE 2015/16 AGAINST 2025/26.

	2015/16		2025/26	
Phrase	% shots	conv.	% shots	conv.
requires a judgement				
"six yard box"	3.22	43.5%	5.96	35.8%
"following a fast break"	0.73	49.3%	3.30	28.4%
does not				
"centre of the box"	32.84	14.2%	36.44	13.2%
"penalty"	0.94	83.1%	1.01	82.8%

The control phrases are as informative as the two that move. "Penalty" admits no interpretive latitude and is unchanged, at 0.94% of shots converting at 83.1% in 2015/16 against 1.01% at 82.8% in 2025/26. "Centre of the box" is nearly as stable. The shift is therefore not a general artefact of the parser or of corpus composition, but is confined to phrases whose application requires a judgement by the annotator.

Scored with the primary fit, the sign of the calibration error reverses across the decade, from -2.7% on 2015/16 to $+11.9\%$ on 2025/26. The estimation sample lies between the two.

C. Re-estimation does not correct the level

Training window	Over-estimate
all three seasons *(primary)*	$+11.9\%$
last two seasons	$+11.9\%$
last season only	$+9.1\%$
recency weighted, half-life one season	$+9.9\%$

The most favourable of these recovers 2.8 of the 11.9 percentage points at the cost of two thirds of the estimation sample. The limitation is not the age of the data. The conversion rate of the forthcoming season is not present in the current season's data under any weighting scheme, so no re-weighting of past observations can recover it.

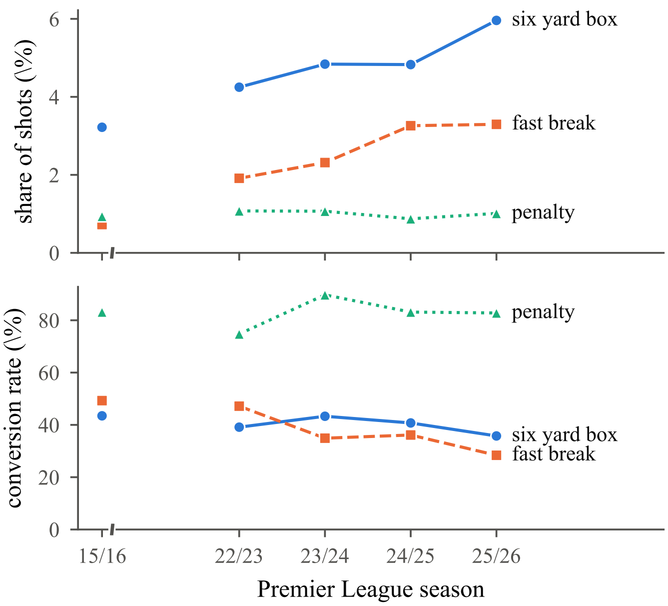

Fig. 3. Share of shots (top) and conversion rate (bottom) for three commentary phrases across the five Premier League seasons in the corpus. The two phrases requiring a judgement move in opposite directions on the two panels, becoming more frequent and less productive; "penalty", which admits no latitude, is flat on both. The axis is broken because 2016/17 to 2021/22 are not in the corpus.

D. In-season intercept correction

Once several hundred shots of a season have been played, the realised conversion rate for that season is observable, and a single additive shift on the intercept restores the level. We walk forward through 2025/26 in blocks of 500 shots, obtaining the shift by bisection on only those shots played before each block, so that no future observation enters the correction applied to it.

	mean xG	actual	over	log loss
uncorrected	0.1267	0.1132	+12.0%	0.29523
recalibrated	0.1165	0.1132	+2.9%	0.29443

The comparison covers 8,694 shots evaluated after a warm-up of 500 shots, approximately twenty matches, during which the model runs uncorrected because the realised rate is too imprecise to correct towards. An additive shift on the intercept is a monotone transformation of the predicted probabilities, so AUC is invariant by construction; the accompanying test suite asserts this. The purpose of that assertion is to prevent a subsequent adjustment to the coefficients, which would improve calibration at the expense of the induced ranking.

E. Relation to prior results on staleness

Robberechts and Davis [4] treat calibration as the primary objective in xG evaluation and report that staleness has a negligible effect. We reproduce their ranking result, finding that a decade-stale model costs 0.0068 AUC, while observing the predicted level move by approximately 10% over the same interval. The two findings are consistent, and taken together they separate two properties that are often reported jointly. Event-data xG is stable in precisely the respect, calibration, in which a text-derived model is not, and the instability we observe is confined to phrases requiring annotator judgement. Coordinates are measurements and do not drift; phrases are authored and do.

We are not aware of an existing characterisation of this effect in text-derived sports features, but we did not search the concept-drift literature systematically for it and make no priority claim. The practical implication stands regardless: a system conditioned on commentary phrasing requires periodic correction of the level within a season, and does not require re-estimation.

VIII. Limitations

The headline comparison rests on a single reference model evaluated on a single season, and this is a limitation of scope rather than of availability. StatsBomb's open releases include six further seasons that ESPN also cover, as listed in Section III, so replicating the comparison on a second league and a second era requires only further collection and would strengthen the result materially. We report one season because that is what we collected, not because it is all there is. A second *provider* is a different matter: no other vendor publishes per-shot xG openly, so the reference model here is necessarily StatsBomb's. The provider-agreement comparison in Section VI-C is subject to the aggregation mismatch described there, and rests on a figure published in a practitioner blog rather than a peer-reviewed source.

The feature set is incomplete in a way we chose rather than inherited. The phrase "from a free kick" converted 60 of 60 in our corpus, which is a leak rather than a finding, and we removed the feature instead of repairing it. A specification that separated the free kick from its outcome would very likely perform better, and the 90.6% figure should be read as a lower bound in that respect. Relatedly, a small number of fixtures return an empty commentary feed and contribute nothing to the corpus; the rate is low but non-zero and is not currently instrumented, so we cannot report it.

We assert that commentary is published quickly enough for live estimation and have no measurement to support it. The obvious shortcut does not work. Each archived commentary item carries a `wallclock` field, and if that recorded publication time then the 3,577 stored matches would already contain the answer. It does not: within a single match the residual between that field and the match clock has a median standard deviation of 0.52 s across 250 matches, with 98.8% of matches below 2 s and a median of three distinct integer values per match. That is the signature of second-level rounding on a constant, so the field is the event time reconstructed as the actual kickoff plus the match clock, and the constant is the kickoff delay, whose median is 60 s. Publication time is not retained anywhere, so the measurement requires live observation, and the statistic must be the minimum of the observed delay across polls rather than its mean, since any single reading also contains the interval since the last event. A scheduled observer exists but on the day tested the cloud scheduler triggered 2 of 12 intervals and did not cover the kickoff. The claim remains untested and is not relied on for any result reported here.

Finally, the novelty claim rests on an incomplete search. Our search of arXiv, OpenAlex and Crossref recovered 5 of 6 probe works known in advance, and not consistently the same five across runs. A subsequent search of grey literature, which located [9] and thereby changed what this paper claims, recovered 3 of 4 probes by name, and one major discussion forum refuses automated access, which is a plausible location for unpublished work of this kind. Both searches are recorded query by query in the accompanying repository. We have not searched licensed indexes, thesis repositories, or non-English sources, and a reader should treat the priority claim in Section II accordingly.

IX. Conclusion and Future Work

A model estimated from commentary sentences recovers 90.6% of a coordinate expected-goals model's discrimination above chance, with a bootstrap interval of $[86.2, 95.0]$, and agrees with it at the team-match level to approximately the degree that one commercial provider agrees with another. The residual gap is not attributable to the extraction step, since richer readers of the same sentence perform worse; it corresponds to information the sentence does not encode, and Section VI-F shows it concentrated in phrases that describe a chance qualitatively rather than geometrically. The model transfers to five further competitions without re-estimation, which we attribute to the commentary template being constructed identically across them.

The failure mode we identify lies in the calibrated level rather than the ranking. Text-derived features drift where coordinate features do not, and the drift is confined to phrases whose application requires a judgement by the annotator, while unambiguous phrases are unchanged over a decade. A single intercept, refitted within the season on shots already played, corrects the level without affecting the ranking. That the same phrases carry both the largest disagreement with the coordinates and the largest movement across seasons suggests these are two consequences of one property of the feature rather than two separate problems.

Three directions follow. First, the drift correction here is a single global intercept, which is the coarsest available remedy; a per-phrase correction that shrinks the coefficient on a term in proportion to the instability of its recent usage would target the mechanism rather than its aggregate effect, and could be estimated within a season by the same procedure. Second, the commentary template is Opta's, and this paper cannot separate what is true of commentary in general from what is true of one provider's house style; a second feed, or a non-English one, would establish which. Third, the result suggests a general question about text-derived features beyond football: wherever a model reads a human-authored description, the descriptive vocabulary is itself a non-stationary instrument, and the separation of stable ranking from drifting level may be the appropriate way to report such models.

StatsBomb's open releases, the largest such collection, cover 25 competitions; there are several hundred professional leagues. For a competition outside that set, and with no prospect of entering it, the practical consequence is the difference between an available xG estimate and none.

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Ethics Statement

The commentary corpus is collected from a public endpoint that requires no authentication and serves text already published to readers of the site. Requests were issued serially with a single connection and no attempt was made to circumvent rate limiting or access controls. The data contains the names of professional footballers acting in their professional capacity, which are already a matter of public record, and no other personal information; no attempt is made to infer anything about individuals beyond the outcome of a shot. The corpus is not redistributed, and the accompanying repository contains the collection code rather than the collected text.

We note one asymmetry worth stating plainly. The argument of this paper is that a free text feed substitutes for a licensed data product. That is advantageous for researchers and practitioners without budgets, which is the motivation, but the feed exists because a commercial provider constructs it, and the substitution is not costless to whoever paid for the annotation.

Use of AI Assistants

AI coding assistants were used throughout this work: for implementation, for exploratory analysis, for literature and grey-literature search, and for drafting and revising this manuscript. All quantitative results were produced by code in the accompanying repository and verified against its test suite rather than reported from model output, and every citation was checked against the arXiv or Crossref API before inclusion. The author is responsible for the content, including the claims, the limitations, and any remaining errors.

Reproducibility

Every quantity reported in this paper is produced by a single script in the accompanying repository [5], which executes the pipeline in dependency order from collection through to the final tables and figures. The figures read their values from the joined data at render time rather than carrying them as literals. The repository also contains 39 test functions, which pytest expands over their parametrised inputs to 91 cases; one of them checks that every figure in this paper is still the figure the artefact holds. These include behavioural guards against outcome leakage [18], a check that the walk-forward correction of Section VII uses no observation from after the block it is applied to, an assertion that the intercept shift leaves AUC unchanged, and a dependency graph that fails if any derived artefact predates the inputs from which it was built. The literature and grey-literature searches on which the priority claim of Section II rests are recorded query by query, together with the recall probes reported in the limitations.

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Appendix A
Extraction

Fields are set by whitelist regular expression over the stripped sentence. Every pattern is listed in `src/shots.py` in the accompanying repository; the substantive design decisions are these.

Only spans matching a chance-describing pattern are retained, and everything else in the sentence is discarded. The retained families are the location phrases (*from the centre of the box*, *from the left side of the box*, *from outside the box*, *from very close range*, *from a difficult angle*, *from long range*), the body-part phrases (*right footed shot*, *left footed shot*, *header*), the assist phrases (*assisted by …with a cross*, *…with a through ball*, *…with a headed pass*, *…with an aerial pass*), and the pattern-of-play phrases (*following a corner*, *following a fast break*, *following a set piece situation*).

Nine fields describe the shot itself rather than its build-up: `header`, `left_foot`, `right_foot`, `six_yard`, `centre_box`, `side_box`, `outside_box`, `long_range` and `difficult_ang`. These are read only from the text preceding the token `assisted by`, which is the repair for leak (4) of Section IV-A: the phrase "headed pass" describes the assist and had been setting the `header` indicator on 1,636 shots.

Free kicks are absent, as described in the limitations.

Appendix B
Estimation

The primary model is a scikit-learn pipeline of `StandardScaler` followed by `LogisticRegression` with `max_iter=2000` and the default ℓ_2 penalty at $C = 1$. Standardisation matters only because `minute` runs from 0 to 95 while the other seventeen fields are binary; it does not change the fit materially. No hyperparameter search was performed and none of the reported numbers depends on one.

The comparison against gradient-boosted trees in Table V uses XGBoost [13] at its defaults. The TF-IDF baseline uses 1–4 grams with `min_df=5` and sublinear term frequency; the embedding baseline uses `all-MiniLM-L6-v2` [14].

The in-season correction of Section VII solves for the additive intercept shift δ satisfying

$$\frac{1}{|B|}\sum_{i\in B}\sigma(z_i+\delta)=\frac{1}{|B|}\sum_{i\in B}y_i, \tag{3}$$

where z_i is the model's logit for shot i, σ the logistic function, and B the set of shots played before the block being scored. The left side is monotone in δ, so it is solved by 60 bisections on $[-5, 5]$, which is well inside floating-point precision. Blocks are 500 shots and the warm-up is 500 shots.

Appendix C
Phrase drift, all seasons

Table IX gives the season-by-season values summarised in Figure 3.

TABLE IX

SHARE OF SHOTS AND CONVERSION RATE BY SEASON, PREMIER LEAGUE. THE CORPUS CONTAINS 2015/16 AND 2022/23 ONWARDS.

	2015/16	2022/23	2023/24	2024/25	2025/26
share of shots (%)					
six yard box	3.22	4.25	4.84	4.83	5.96
following a fast break	0.73	1.91	2.31	3.26	3.30
centre of the box	32.84	36.56	36.07	35.86	36.44
penalty	0.94	1.08	1.07	0.87	1.01
conversion rate (%)					
six yard box	43.5	39.1	43.3	40.7	35.8
following a fast break	49.3	47.2	34.9	36.1	28.4
centre of the box	14.2	14.1	15.6	14.5	13.2
penalty	83.1	74.7	89.7	83.1	82.8
shots	9,502	9,205	10,022	9,508	9,194
overall conversion (%)	10.5	11.8	12.4	11.7	11.3